

AI-BFSI Internship Project Documentation

Project Planning Template – Subscription Waste Detector

Executive Summary

Effective project planning provides a structured roadmap for developing software within the available time and resources. It defines objectives, phases, deliverables, risks, and priorities before coding begins.

The Subscription Waste Detector was planned using a **phased development approach**. The project was divided into manageable stages: problem analysis, requirement gathering, solution design, implementation, testing, and documentation. This structured planning minimized complexity while ensuring alignment with project objectives.

Project Overview

Attribute	Details
Project Name	Subscription Waste Detector
Project Type	AI-Assisted Financial Analysis Web Application
Domain	Banking, Financial Services and Insurance (BFSI)
Development Approach	Phased Software Development
Platform	Web Application
Primary Goal	Detect recurring subscriptions and provide AI-powered financial insights

Project Objectives

- Develop a practical financial analysis application.
- Detect recurring subscription payments.
- Improve financial awareness.
- Visualize subscription spending.
- Generate AI-powered recommendations.
- Deliver a functional and maintainable software solution.

Project Scope

Included

- CSV file upload
- Excel file upload
- Manual expense entry
- Subscription detection

- Dashboard visualization
- Plotly charts
- Renewal reminders
- AI-generated recommendations using Groq API

Excluded

- User authentication
- Database integration
- Cloud synchronization
- Bank API integration
- Email notifications
- Mobile application
- Multi-user support
- Investment management
- Tax planning

Project Development Phases

Phase 1 – Problem Identification

- Study recurring subscription challenges.
- Analyze existing financial management practices.
- Define the problem statement.
- Evaluate possible solution approaches. **Deliverable:** Approved project concept.

Phase 2 – Requirement Analysis

- Identify functional requirements.
- Define project scope.
- Determine expected outputs.
- Select technologies. **Deliverable:** Requirements specification.

Phase 3 – Solution Design

- Design overall architecture.
- Plan workflow.
- Define data processing approach.
- Design dashboard layout. **Deliverable:** Solution architecture documentation.

Phase 4 – Development

- Build Streamlit interface.
- Implement file upload functionality.
- Develop subscription detection logic.
- Create Plotly visualizations.
- Integrate Groq API.
- Assemble dashboard components. **Deliverable:** Working prototype.

Phase 5 – Testing

- Verify file upload functionality.
- Validate subscription detection.
- Test dashboard visualization.
- Verify AI recommendation generation.
- Check UI consistency. **Deliverable:** Validated application.

Phase 6 – Documentation

- Prepare technical documentation.
- Document architecture and workflow.
- Compile project reports. **Deliverable:** Complete documentation set.

Work Breakdown Structure (WBS)

Phase	Major Tasks
Planning	Problem identification, objective definition
Analysis	Requirement gathering, scope definition
Design	Architecture and workflow design
Development	Feature implementation
Testing	Functional validation
Documentation	Technical reports and project documentation

Project Timeline

Phase	Expected Outcome
Planning	Project concept finalized
Analysis	Requirements completed
Design	Architecture prepared
Development	Functional application developed
Testing	Features verified
Documentation	Final reports completed

Resource Planning

Software

- Visual Studio Code
- Python
- Streamlit

Libraries

- Pandas
- Plotly
- OpenPyXL

- Groq SDK

Hardware

- Personal computer for development and testing

Risk Assessment

Risk	Mitigation Strategy
Poor-quality expense data	Validate uploaded files before processing
Incorrect recurring detection	Test with multiple datasets
AI response inconsistency	Present AI output as recommendations rather than financial advice
Unexpected data formats	Implement preprocessing and validation

Quality Planning

Quality was maintained by:

- Testing uploaded datasets
- Verifying subscription detection
- Checking dashboard accuracy
- Validating AI recommendations
- Reviewing documentation consistency

Expected Deliverables

- Functional Streamlit application
- Subscription detection module
- Interactive dashboard
- AI recommendation module
- Technical documentation
- Project reports

Success Criteria

The project is successful when:

- Users can upload expense files successfully.
- Recurring subscriptions are detected correctly.
- Monthly subscription spending is visualized.
- AI recommendations are generated successfully.
- The application provides meaningful financial insights.

Future Planning Considerations

Future versions may include:

- User authentication
- Database support
- Cloud deployment
- Bank API integration
- Email notifications
- Multi-user functionality
- Advanced financial analytics

Conclusion

The Project Planning Template provides a structured roadmap for the successful development of the Subscription Waste Detector. By dividing the project into phases, identifying scope, allocating resources, and assessing risks, the planning process ensured an organized and manageable development lifecycle.